Dynamic and Asymmetric Enhancement for Remote Sensing Image Change Captioning

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***Abstract*—Remote sensing image change captioning (RSICC) plays a critical role in automated environmental monitoring by generating natural language descriptions that provide intuitive interpretations of changes between bi-temporal remote sensing images. Despite recent advancements, existing methods suffer from two fundamental limitations: 1) static encoding architectures fail to account for scene complexity and semantic diversity during feature extraction, leading to inflexible representation learning; and 2) symmetric computational structures are inherently unsuitable for modeling the asymmetric temporal dependencies inherent in "before-to-after" image pairs. To address these challenges, we propose a dynamic asymmetric encoder (DAE), which introduces two key innovations. First, we design a difference-guided dynamic convolution (DGDC) module that adaptively adjusts convolutional parameters using input-driven scaling factors and offsets, thereby enabling scene-aware intra-image feature enhancement. Second, we develop a multiexpert temporal interaction (METI) module that establishes an asymmetric computational topology: the "after" image branch actively perceives change information relative to the "before" image through three heterogeneous interaction experts, followed by feature fusion. This design allocates greater computational capacity to the "after" image branch while preserving temporal coherence. Furthermore, we introduce a multilevel feature aggregator (MFA) that enhances the representation of salient changed regions across multiple scales via an iterative reinforcement mechanism. Experimental results validate the effectiveness of the proposed method, demonstrating state-of-the-art performance on two benchmark RSICC datasets. We publicly release our code repository at https://github.com/TTXiann/Dynamic-Asymmetric to facilitate future research.**

***Index Terms*—Asymmetric structure, change captioning, dynamic encoding, multilayer refinement, remote sensing.**

I. Introduction

CHANGES on Earth are driven by both human activities—such as urbanization—and natural disasters. With the rapid advancement of Earth observation technologies, multitemporal remote sensing imagery has become increasingly available, providing rich data resources for monitoring and understanding environmental dynamics. This progress has also spurred the development of intelligent analysis techniques. Remote sensing image change captioning (RSICC) aims to describe semantic changes between bi-temporal images captured over the same geographic area using natural language. This task enables automatic perception and interpretation of surface changes and finds broad applications in disaster assessment, urban planning, land use monitoring, and infrastructure development [1], [2], [3].

While image change captioning (ICC) has seen progress in the domain of natural images [4], [5], [6], the RSICC task poses unique challenges, primarily in two aspects.

1) *Long Temporal Intervals:* Remote sensing images are often acquired months or even years apart, during which lighting conditions and seasonal variations introduce substantial nonsemantic differences.
2) *Feature Ambiguity:* The high-altitude, nadir-view perspective commonly used in remote sensing lacks 3D geometric cues, leading to ambiguous representations of ground objects.

Consequently, directly applying existing natural ICC methods to RSICC often yields suboptimal performance. To address this, dedicated RSICC models have been proposed. For example, RSICCFormer [7] integrates a differential encoding mechanism with a multistage bi-temporal feature interaction module to generate semantically meaningful change descriptions at multiple feature levels. PSNet [8] employs progressive difference-aware layers and a multiscale feature enhancement module to capture change information across scales. PromptCC [9] first uses a pretrained binary change classifier to detect whether a change has occurred and then leverages learnable prompt vectors to guide a language model in generating descriptive captions. Additionally, several works [10], [11], [12] incorporate remote sensing image change detection (RSICD) models, utilizing their pixel-level binary change masks as spatial priors to improve the localization and accuracy of generated descriptions.

Despite these advances, existing methods still exhibit two critical limitations in modeling semantic changes.

1) *Static Encoding Problem:* Current approaches typically rely on fixed, handcrafted visual encoding pipelines to process input images [Fig. 1(a)]. Such static schemes lack adaptability. Intuitively, scenes involving simple versus complex semantic changes—or pseudo-changes

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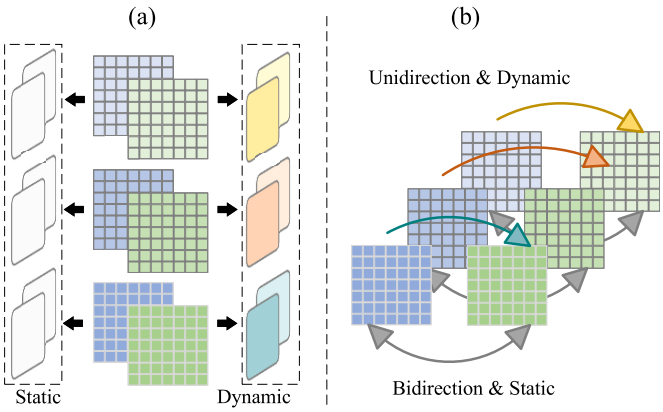


Fig. 1. Comparison between our method and existing methods during (a) intra-image encoding phase and (b) cross-image interaction phase.

versus genuine semantic changes—should be encoded using distinct strategies. Moreover, manually designed pipelines heavily depend on expert knowledge and struggle to generalize across diverse input patterns.

2) *Symmetric Encoding Problem:* Most methods adopt symmetric architectures for encoding bi-temporal images. While reasonable for intra-image feature extraction, this design ignores the inherent temporal asymmetry between the "before" and "after" images during cross-temporal interaction [Fig. 1(b)]. From a causal perspective, change interpretation is naturally anchored in the "after" image, which reflects the outcome of any transformation, whereas the "before" image represents only the initial state. Thus, the "after" image should play the dominant role in perceiving changes, while the "before" image serves as the reference. Explicitly modeling this temporal role distinction can enhance the model's sensitivity to true semantic changes.

Inspired by the aforementioned analysis, we propose a dynamic asymmetric encoder (DAE). The core innovations of the proposed method are twofold: 1) the introduction of a dynamic encoding mechanism that enables input-driven adaptive feature encoding, as illustrated in Fig. 1(a) and (b); and 2) the construction of a heterogeneous bi-temporal phase encoding architecture, which explicitly differentiates the processing pathways for the "before" and "after" images at the computational topology level, as shown in Fig. 1(b). Specifically, each encoding layer in the proposed DAE comprises two sequential phases: intra-image feature enhancement and temporal feature interaction. In the first phase, we design a difference-guided dynamic convolution (DGDC) Module. This module employs multiple convolutional layers with identical architectures and leverages intertemporal difference features as guiding signals to generate scaling factors and offsets for the parameters of each convolutional layer. This enables dynamic adaptation of the convolutional parameters, allowing the model to focus on key local features within regions of interest. In the second phase, we introduce a multiexpert temporal interaction (METI) module, which operates exclusively on the "after" branch. The METI module integrates information from the "before" branch to enable the "after" image to perceive semantic changes while preserving the integrity of temporal information. To further enhance adaptability to diverse input scenarios, we design three heterogeneous interaction experts, each specialized in extracting distinct types of temporal features. A subsequent feature selection module dynamically computes importance weights for the outputs of these experts and performs weighted fusion. Consequently, METI constructs a rich temporal interaction space capable of effectively handling diverse samples. Compared with most existing methods—which typically adopt static and bidirectional encoding architectures—our approach allocates more computational resources to encoding the "after" image, thereby achieving structural asymmetry between the "before" and "after" branches. Furthermore, the extensive use of dynamic encoding modules significantly enhances the model's adaptability to input variations. These design choices collectively facilitate more discriminative representation of change semantics. Additionally, we develop a multilevel feature aggregator (MFA) to integrate multiscale "after" features output by the encoder. The MFA iteratively reinforces salient information while suppressing irrelevant features, yielding semantically rich and structurally coherent change representations. The final change representation is then fed into a standard Transformer decoder to generate accurate change captions.

The main contributions of this article can be summarized as follows.

1) We propose a DAE, which enables input-dependent dynamic intra-image feature enhancement and adaptive unidirectional temporal interaction. This design significantly improves the model's sensitivity to semantic variations across different scenarios and enhances its adaptability to diverse input patterns.
2) We construct an MFA that effectively integrates multiscale change features generated by the asymmetric change encoder, thereby more comprehensively capturing semantic change information within images.
3) Extensive experiments conducted on the LEVIR-CC and DUBAI-CC datasets validate the effectiveness of the proposed method, achieving state-of-the-art performance.

II. Related Works

A. Remote Sensing ICC

ICC seeks to interpret visual differences between two images and describe them in natural language [5], [6], [13], [14], [15], [16]. Although ICC has achieved significant progress on natural and synthetic datasets, RSICC remains underexplored. RSICC presents unique challenges, including semantic noise caused by seasonal, atmospheric, and illumination variations, as well as the lack of discernible 3-D structure due to high-altitude imaging. These factors significantly complicate accurate change recognition and subsequent caption generation.

Two related tasks provide context for RSICC. Image captioning (IC) generates natural language descriptions from a single image by recognizing objects, actions, scenes, and attributes [17], [20], [39], [40]. In contrast, ICC operates

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on pairs of co-registered images captured at different times, requiring not only scene understanding but also change detection (CD)—i.e., identifying what changed, where, and the nature of the change—and translating this information into natural language. RSICD focuses on localizing changed regions via binary masks [21], [22], [23], [24] without capturing the semantic content or temporal evolution of changes. In contrast, RSICC must model both spatial and temporal dependencies to generate accurate natural language descriptions of dynamic changes.

Several models have been proposed for RSICC. Early CNN-based methods [25], [26] were limited by small-scale datasets. The introduction of the LEVIR-CC dataset [7] provided a standardized benchmark for evaluation. More recently, Transformer-based methods have demonstrated superior performance: RSICCformer [7] improves bi-temporal feature difference representation using multistage fusion. ChgCap [7] employs bi-temporal self-attention and residual units to enhance consistent and inconsistent features. PSNet [8] uses progressive difference-aware layers and scale-aware modules for multiscale feature extraction. ICT-Net [27] integrates multiscale features with interactive encoding and cross-gated attention. MFRNet [28] combines multiscale CNN features with joint attention and dense fusion for robust representation. MAF-Net [29] captures multilevel changes through a CNN backbone and a multiscale change-aware encoder. SFT [30] reduces computational cost by limiting attention scope. CTM [31] models temporal dynamics via manifold mapping and Bayesian diffusion to handle illumination and seasonal variation.

Despite these advances, existing methods predominantly employ static and symmetric encoding architectures, which are inherently limited in capturing the diversity and complexity of visual changes. In contrast, we propose a DAE that facilitates adaptive intra-image enhancement combined with unidirectional cross-image interaction. This design introduces a flexible encoding framework and an enriched interaction space, further augmented by an MFA to generate semantically enriched representations. Consequently, the proposed method enhances both the quality and robustness of the generated captions.

B. RSICC With External Knowledge

The integration of external knowledge has been demonstrated to enhance model performance and learning efficiency in various vision tasks [32], [33], [34]. In the context of RSICC, attention-based models often exhibit limitations in accurately localizing changes without explicit guidance. To address this issue, recent methods have incorporated CD modules within RSICC frameworks, leveraging detected change regions as prior information to improve both perception and localization capabilities. For example, Pix4Cap [10] utilizes a pretrained CD model to generate pseudo-labels and introduces an auxiliary CD branch equipped with a semantic fusion module to enhance feature integration. DACC [11] fuses binary CD masks with input images at both pixel and feature levels to emphasize features associated with changes. CD4C [12] employs a multitemporal difference fusion strategy guided by detected changes to highlight foreground variations. PromptCC [9] decouples the RSICC task into classification and generation subtasks, employing GPT-2 alongside visual and learnable prompts to produce more accurate and descriptive captions. SGD-RSCCN [35] integrates syntactic dependencies using SpaCy, constructing a knowledge-driven language model that effectively differentiates between changed objects and reference entities.

Although these methods benefit from incorporating external knowledge, they often introduce additional computational complexity, exhibit limited generalization across diverse scenarios, and remain sensitive to the quality of the provided prior information. In contrast, this work proposes a knowledge-free framework that simplifies the overall architecture, reduces computational overhead, and achieves state-of-the-art performance without reliance on external supervision.

III. Methodology

A. Preliminary

The Transformer [36] consists of multihead attention (MHA) and feedforward network (FFN) blocks. MHA captures token dependencies via scaled dot-product attention

$$\begin{aligned}\mathrm{MHA}(Q,K,V) &= \mathrm{Concat}(\mathrm{head}_1,\ldots,\mathrm{head}_h)\,W^O\\ \mathrm{head}_i &= \mathrm{Attention}\left(QW_i^Q, K\ W_i^K, V\ W_i^V\right)\\ \mathrm{Attention}(Q,K,V) &= \mathrm{Softmax}\left(\frac{QK^\top}{\sqrt{d_h}}\right)V \end{aligned} \tag{1}$$

where $d_h = d/h$, $W_i^{Q,K,V} \in \mathbb{R}^{d\times d_h}$ and $W^O \in \mathbb{R}^{hd_h\times d}$. The FFN applies two linear layers with a GELU activation

$$\mathrm{FFN}(X) = \mathrm{GELU}(XW_1)\,W_2 \tag{2}$$

with $W_1 \in \mathbb{R}^{d\times 4d}$ and $W_2 \in \mathbb{R}^{4d\times d}$.

B. Overview

The overall architecture of the proposed model is illustrated in Fig. 2. It comprises four components: 1) a visual backbone network that extracts features from input images; 2) a DAE designed to capture change cues from bi-temporal image pairs; 3) an MFA that fuses multiscale features generated by the DAE into a unified change representation; and 4) a language model that autoregressively generates semantic descriptions of the detected changes.

Given bi-temporal image inputs I_1 and I_2, we first employ a pretrained visual backbone network to extract their respective visual features, denoted as F_1 and F_2. A linear projection layer is then applied to reduce the channel dimensionality of these features, resulting in $X_1, X_2 \in \mathbb{R}^{d\times h\times w}$. Based on this representation, the DAE module produces a set of refined "after" features $\mathcal{X}_2 = \{X_2^1, X_2^2, \ldots, X_2^L\}$. The MFA subsequently encodes $\mathcal{X}_2$ into a compact final representation X_c. Finally, a language model generates the corresponding change caption conditioned on X_c. The complete computational pipeline of the proposed model can be formally expressed as follows:

$$\begin{aligned} F_1, F_2 &= \mathrm{Backbone}(I_1, I_2)\\ X_1^0, X_2^0 &= \mathrm{Linear}\left(\mathrm{Reshape}(F_1), \mathrm{Reshape}(F_2)\right)\end{aligned}$$

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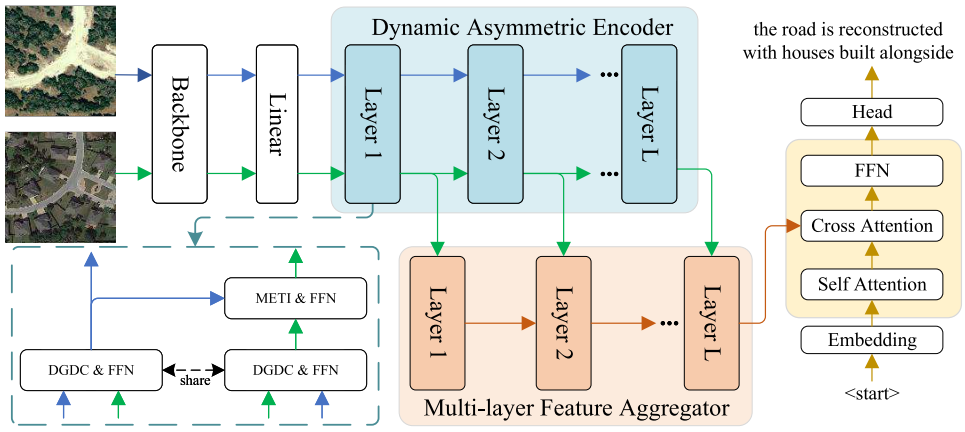


Fig. 2. Overall architecture of the proposed model. The core components include a DAE designed to model semantic changes, which incorporates a DGDC module and an METI module. Additionally, an MFA progressively refines multiscale features into semantically rich change representations.

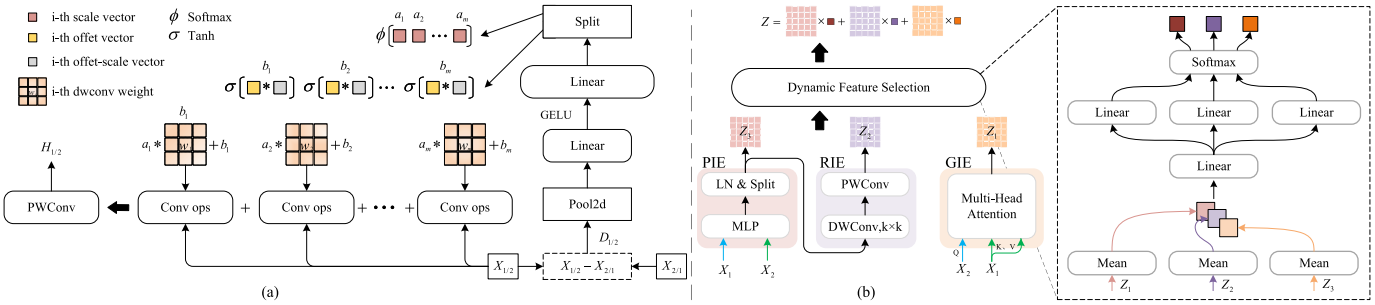


Fig. 3. Architecture of the proposed modules. (a) DGDC module. (b) METI module.

$$\begin{aligned} \mathcal{X}_2 &= \mathrm{DAE}\left(X_1^0, X_2^0\right) \\ X_c &= \mathrm{MFA}\left(\mathcal{X}_2\right) \\ \mathrm{Caption} &= \mathrm{Decoder}\left(X_c\right). \end{aligned} \tag{3}$$

C. Dynamic Asymmetric Encoder

The architecture of the DAE is illustrated in Fig. 2 (blue box), which comprises two main stages: *intra-image feature enhancement*, which refines image features to highlight semantically meaningful regions; *temporal feature interaction*, which allows the "after" image to attend to the content of the "before" image, effectively capturing change-related semantics while preserving temporal fidelity.

Based on this design, the computational process within each encoding layer can be formalized as follows:

$$\begin{aligned} \hat{X}_1^l, \hat{X}_2^l &= \mathrm{DGDC}\left(X_{1/2}^l, X_{2/1}^l\right) \\ \bar{X}_2^l &= \mathrm{METI}\left(\hat{X}_2^l, \hat{X}_1^l\right). \end{aligned} \tag{4}$$

Note that DGDC is applied to both X_1^l and X_2^l, while METI refines the representation of $\hat{X}_2^l$ with $\hat{X}_1^l$ as reference; therefore, we set $X_1^{l+1} = \hat{X}_1^l$ and $X_2^{l+1} = \bar{X}_2^l$.

D. DGDC Layer

The DGDC module, illustrated in Fig. 3(a), adopts a predefined-adjustment strategy rather than directly generating convolutional kernels. The module comprises m convolutional layers with identical architectures but independent parameters, enabling the extraction of local features from multiple perspectives. To facilitate input-driven dynamic convolutions, a lightweight network predicts scaling matrices $\mathcal{A}$ and offset matrices $\mathcal{B}$ for each layer. These matrices adaptively adjust the convolutional parameters based on the input, thereby enhancing the model's adaptability. Standard convolution and self-attention mechanisms often struggle to capture differential information, which depends on comparative relationships between images. To address this limitation, we compute relative difference features between bi-temporal images using subtraction and utilize these as guiding signals to generate $\mathcal{A}$ and $\mathcal{B}$. This design focuses feature extraction on regions of change, improving the model's sensitivity to semantic differences.

Given the input features X_1, X_2, we first compute the difference features as follows:

$$D_{1/2} = X_{1/2} - X_{2/1}. \tag{5}$$

We then employ a multilayer perceptron (MLP) to generate the initial scaling and offset matrices as follows:

$$\begin{aligned} \hat{D}_{1/2} &= \mathrm{MLP}\left(\mathrm{AvgPool}_{2\mathrm{d}}\left(D_{1/2}\right)\right) \\ \hat{A}, \hat{B} &= \mathrm{Split}\left(\hat{D}_{1/2}\right) \end{aligned} \tag{6}$$

where MLP denotes a pipeline consisting of Linear $\rightarrow$ GELU $\rightarrow$ Linear layers. Next, we apply the Softmax function to the scaling matrix to compute normalized scores for each depth-wise convolutional layer (DWC), i.e., $A = \mathrm{Softmax}(\hat{A})$. For the offset matrix, we utilize the Tanh function to transform

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each element within the range $[-1, 1]$, i.e., $B = \text{Tanh}(\hat{B} \odot C)$. $C \in \mathbb{R}^{m\times d}$ is a trainable offset-scaling matrix, initialized to 0.1, used to prevent matrix $\mathcal{B}$ from saturating prematurely. Let the weights of m independent DWC layers be denoted as $[w_1, \ldots, w_m]$, where each $w_i \in \mathbb{R}^{d\times 1\times k\times k}$. At the lth layer, with kernel size $k = 2l+1$. These parameters are adaptively adjusted using the obtained scaling matrix $A = \{a_1, a_2, \ldots, a_m\} \in \mathbb{R}^{m\times d}$ and offset matrix $B = \{b_1, b_2, \ldots, b_m\} \in \mathbb{R}^{m\times d}$ as follows:

$$\bar{w}_i = a_i \cdot w_i + b_i. \tag{7}$$

Finally, we employ DWCs with adaptively adjusted weights to encode the input features. The output feature maps from the m DWC layers are element-wise summed and subsequently passed through a pointwise convolution (PWConv) to facilitate cross-channel feature mixing, as illustrated in the following:

$$H_{1/2} = \text{PWC}\left(\sum_{i=1}^{m} \text{DWC}_{\bar{w}_i}\left(X_{1/2}\right)\right). \tag{8}$$

The proposed DGDC method is characterized by two key attributes: 1) multigroup isomorphic convolutions that extract features from multiple perspectives, thereby enhancing local modeling capabilities; and 2) dynamic convolution, which captures differential information between bi-temporal features to guide each input in focusing on relevant regions.

Following the standard token- and channel-mixing paradigm, we incorporate layer normalization (LN) [37] and a FFN [36]. The final output of the DGDC layer can be expressed as follows:

$$\begin{aligned} \hat{H}_{1/2} &= \text{LN}\left(X_{1/2} + H_{1/2}\right) \\ \hat{X}_{1/2} &= \text{LN}\left(\hat{H}_{1/2} + \text{FFN}\left(\hat{H}_{1/2}\right)\right). \end{aligned} \tag{9}$$

E. METI Layer

First, we propose an asymmetric feature processing mechanism that utilizes the "before" features as a reference to optimize and update the "after" features, thereby enabling efficient change perception. Building upon this, we design three heterogeneous interaction modules to model feature interactions at different spatial scales: the global interaction expert (GIE), the region interaction expert (RIE), and the point interaction expert (PIE). Furthermore, we introduce a dynamic feature selection (DFS) mechanism to adaptively fuse the outputs of each expert, generating a unified and discriminative feature representation. The structure of the METI module is illustrated in Fig. 3(b).

1) Global Interaction Expert: The GIE is implemented based on the MHA mechanism. Specifically, we utilize the "after" feature X_2 as the query and the "before" feature X_1 as the key–value pair. By computing the dot-product similarity between the query and the key, each token in X_2 can attend to the entire X_1. This process can be formulated as follows:

$$Z_1 = \text{MHA}\left(Q = X_2, K = X_1, V = X_1\right). \tag{10}$$

2) Region Interaction Expert: The RIE is implemented based on depthwise separable convolutions. Specifically, we first apply an MLP to mix and compress the bi-temporal features across channels. Subsequently, the processed features are fed into a convolutional layer to achieve region-level interactions. This process is formulated as follows:

$$Z_2 = \text{PWC}\left(\text{DWC}_k\left(\text{MLP}\left([X_2; X_1]\right)\right)\right) \tag{11}$$

where $[\cdot;\cdot]$ represents concatenation along the channel dimension. At the lth layer, the kernel size is set to $k = 2l + 1$ to construct multiscale regional interactions. The MLP is defined as Linear → GELU → Linear → LayerNorm.

3) Point Interaction Expert: The PIE is implemented based on MLP. This MLP performs token-level channel mixing and compression on bi-temporal features as follows:

$$Z_3 = \text{MLP}\left([X_2; X_1]\right). \tag{12}$$

For parameter efficiency, the MLP used in the PIE shares some parameters with the MLP employed in the RIE.

4) Dynamic Feature Selection: The three heterogeneous experts generate interaction features of different types. Considering that different scenarios may require different interaction patterns, we construct the DFS module, which aims to dynamically compute the importance of each expert's output and combine them accordingly. Specifically, we first compute a global vector $Z_g \in \mathbb{R}^d$ as follows:

$$Z_g = \text{Linear}\left(\left[\text{Avg}\left(Z_1\right); \text{Avg}\left(Z_2\right); \text{Avg}\left(Z_3\right)\right]\right). \tag{13}$$

Next, we predict scores $S \in \mathbb{R}^{3\times d}$ for each expert using three linear layers, followed by normalization via the Softmax function:

$$S = \text{Softmax}\left(\left[\text{Linear}\left(Z_g\right); \text{Linear}\left(Z_g\right); \text{Linear}\left(Z_g\right)\right]\right). \tag{14}$$

Finally, we compute a weighted sum of the input expert features based on the obtained scores to derive a unified feature

$$Z = \text{Sum}\left(S \cdot [Z_1; Z_2; Z_3]\right). \tag{15}$$

Overall, the expert models employ distinct computational paradigms and parameter scales, thereby constructing diverse cross-image temporal interaction mechanisms. Additionally, the DFS module further establishes a flexible temporal interaction space by computing weight scores for each expert and performing weighted fusion. The aforementioned design significantly enhances the model's capacity to represent complex semantic changes.

Similar to the DGDC layer, we incorporate LN and FFN within the METI layer. The final output of the METI layer can be expressed as follows:

$$\begin{aligned} \bar{Z} &= \text{LN}\left(X_2 + Z\right) \\ \bar{X}_2 &= \text{LN}\left(\bar{Z} + \text{FFN}\left(\bar{Z}\right)\right). \end{aligned} \tag{16}$$

F. Multilevel Feature Aggregation

The DAE module comprises multiple encoding layers characterized by a dynamic and asymmetric design, enabling each layer to generate discriminative features. To extract meaningful semantics while mitigating noise, we introduce the MFA module, which progressively refines change semantics from multilevel features, resulting in a fine-grained representation. The structure of the MFA module is illustrated in Fig. 4. Specifically, at each level, a convolutional block (CB) is

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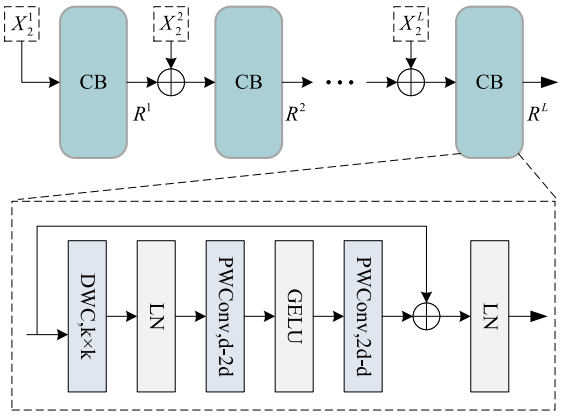


Fig. 4. Structure of the proposed MFA.

employed to encode the input features X_2^l, thereby enhancing local feature dependencies. Simultaneously, the input features from the subsequent level are integrated with the output features of the current level through an element-wise addition operation, facilitating cross-level information propagation and integration. The computation for the lth MFA layer can be expressed as follows:

$$R^{l+1} = \begin{cases} \mathrm{LN}\left(\mathrm{CB}^l\left(X_2^l\right)\right), & l = 0 \\ \mathrm{LN}\left(\mathrm{CB}^l\left(R^l + X_2^l\right)\right), & l > 0. \end{cases} \tag{17}$$

We adopt the ConvNeXt Block [38] as the CB and replace its fixed 7×7 DWC with increasing kernel sizes, defined as $k = 2l + 1$. This multiscale convolution design aligns with the overarching philosophy of other modules in this work, ensuring consistent and robust feature representation across varying scales. Finally, the output of the MFA module, denoted R^L, serves as the change representation X_c, which is subsequently fed into the language model for caption generation.

G. Training and Inference

We employ a standard Transformer decoder [36] with L_{dec} layers as the text decoder and use the cross-entropy loss to measure the discrepancy between the generated caption and the ground-truth word sequence $w^* = \{w_1^*, w_2^*, \ldots, w_T^*\}$. The training objective is to minimize the caption generation loss, defined as follows:

$$\mathcal{L} = -\sum_{t=1}^{T} \log\left(p\left(w_t^* \left| w_{<t}^*\right.\right)\right). \tag{18}$$

We propose a two-stage training protocol. In Stage 1, the visual backbone is frozen. All other parameters are optimized to learn stable interaction patterns on top of the fixed visual features. In Stage 2, only the DFS module is unfrozen. To encourage the discovery of richer interaction patterns, we inject a layer-dependent scaling factor into (14) as follows:

$$S = \mathrm{Softmax}\left(\left[\mathrm{Linear}\left(Z_g\right); \mathrm{Linear}\left(Z_g\right); \mathrm{Linear}\left(Z_g\right)\right] \cdot \sqrt{l}\right)$$

where l denotes the encoder layer index.

During inference, sentences are generated word-by-word using a beam search strategy with a beamwidth of 3.

IV. Experimental Results and Analyses

A. Datasets

The **LEVIR-CC dataset** extends a building CD dataset originally comprising 637 bi-temporal, very high-resolution (0.5 m/pixel) satellite image pairs from 20 regions in Texas, USA, with a temporal span of 5–14 years. For change captioning, the images were divided into 10 077 smaller bi-temporal tiles of size 256×256 pixels. Each tile is annotated for the presence or absence of change, yielding 5038 changed and 5039 unchanged pairs. Each pair is associated with five descriptive sentences detailing the observed changes. Following [7], the dataset is split into 6815 training, 1333 validation, and 1929 test pairs.

The **DUBAI-CC dataset**, introduced by Hoxha et al. [26], provides a comprehensive characterization of urbanization dynamics in the Dubai region from 2000 to 2010. It is constructed from Landsat 7 ETM+ multispectral imagery with a spatial resolution of 30 m/pixel. The dataset contains 500 image pairs, each of size 50×50 pixels, with each pair associated with five descriptive sentences, yielding a total of 2500 text descriptions. Following the original protocol [26], the dataset is partitioned into training, validation, and test splits containing 300, 50, and 150 image pairs, respectively.

B. Experimental Setup

1) Implementation Details: We employ a frozen, pretrained ResNet-101 (comprising only the first three stages) as the visual encoder to extract features from input image pairs. The feature dimension d is set to 512 and L_{dec} is set to 1. On the LEVIR-CC and DUBAI-CC datasets, the total batch sizes are set to 40 and 24, respectively, with the number of training epochs set to 10 and 20, respectively. We use the Adam optimizer [41] with an initial learning rate of 1×10^{-4}, and the learning rate is decayed using a cosine annealing schedule. During training, model performance is evaluated on the validation set after each epoch, and training is terminated early if the BLEU-4 score does not improve for three consecutive epochs. All experiments were conducted on a server equipped with an Intel Xeon Gold 6430 CPU and two NVIDIA RTX 4090 GPUs.

2) Evaluation Metrics: We employed several widely adopted automatic evaluation metrics, including BLEU-N ($N = 1, 2, 3, 4$) [42], ROUGE$_L$ [43], METEOR [44], and CIDEr [45]. These metrics are utilized to quantify the degree of match between the model-generated sentences and the human-annotated reference sentences. A higher score indicates better quality. Besides, we adopt an average metric

$$S_m^* = (\text{BLEU-4} + \text{ROUGE}_L + \text{METEOR} + \text{CIDEr})/4. \tag{19}$$

C. Comparison With State-of-the-Art Methods

In this section, we conduct a comprehensive comparative evaluation of the proposed method against state-of-the-art approaches on two RSICC datasets: LEVIR-CC and DUBAI-CC. The compared open-source methods include DUDA [4], MCCFormer [2], RSICCFormer [7], Chg2Cap [46], Prompt-CC [9], SFT [30], SEN [47],

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TABLE I

PERFORMANCE COMPARISON WITH STATE-OF-THE-ART METHODS ON THE LEVIR-CC DATASET. THE BEST AND SECOND-BEST RESULTS ARE INDICATED IN RED AND BLUE, RESPECTIVELY. † DENOTES OPEN-SOURCE METHODS

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE$_L$	CIDEr	S_m^*
SGD-RSCNN	84.17	75.16	68.05	62.48	39.18	74.24	136.27	78.04
CD4C	86.74	78.71	71.48	65.41	40.56	75.72	138.00	79.92
CTM	86.78	79.24	72.65	65.45	40.63	75.72	138.78	80.15
DUDA†	81.44	72.22	64.24	57.79	37.15	71.04	124.32	72.58
MCCFormers-S†	79.80	70.26	62.68	56.68	36.17	69.46	120.39	70.68
MCCFormers-D†	80.42	70.87	62.86	56.38	37.29	70.32	124.44	72.11
PSNet†	83.86	75.13	67.89	62.11	38.80	73.60	132.62	76.78
RSICCformer†	84.72	76.27	68.87	62.77	39.61	74.12	134.12	77.66
Prompt-CC†	83.66	75.73	69.10	63.54	38.82	73.72	136.44	78.13
SEN†	85.10	77.05	70.01	64.09	39.59	74.57	136.02	78.57
SFT†	84.56	75.87	68.64	62.87	39.93	74.69	137.05	78.64
Diffusion-RSCC†	-	-	-	60.90	37.80	71.50	125.60	73.95
Chg2Cap†	86.14	78.08	70.66	64.39	40.03	75.12	136.61	79.04
TACC†	85.49	77.41	70.52	64.62	40.07	74.96	137.17	79.21
Ours	87.13	79.71	73.36	68.12	41.29	76.53	143.24	82.30
Ours (FT)	87.40	79.99	73.71	68.52	41.40	76.53	144.01	82.62

TABLE II

PERFORMANCE COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DUBAI-CC DATASET. THE BEST AND SECOND-BEST RESULTS ARE INDICATED IN RED AND BLUE, RESPECTIVELY. † DENOTES OPEN-SOURCE METHODS

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE$_L$	CIDEr	S_m^*
SGD-RSCCN	77.15	66.80	58.07	50.27	34.07	65.88	118.42	67.16
CD4C	74.13	63.11	53.13	43.55	30.27	60.25	108.66	60.68
DUDA†	58.82	43.59	33.63	25.39	22.05	48.34	62.78	39.64
MCCFormers-S†	52.97	37.02	27.62	22.57	18.64	43.29	53.81	34.58
MCCFormers-D†	64.65	50.45	39.36	29.48	25.09	51.27	66.51	43.09
RSICCformer†	67.92	53.61	41.37	31.28	25.41	51.96	66.54	43.80
SFT†	67.30	55.97	47.00	37.30	26.32	56.38	91.59	52.90
Diffusion-RSCC†	-	-	-	33.30	27.40	56.50	88.70	51.48
Ch2Cap†	72.04	60.18	50.84	41.70	28.92	58.66	92.49	55.44
TACC†	72.17	58.65	48.24	38.59	28.17	58.93	95.45	55.28
Ours	75.98	64.48	54.68	44.45	32.01	67.35	112.14	63.99
Ours (FT)	76.08	64.77	54.95	44.76	32.38	68.06	113.94	64.78

Diffusion-RSCC [48], and TACC [49]; while the closed-source methods include SGD-RSCCN [35], CD4C [12], and CTM [31].

As shown in Table I, our model achieves the best performance across all evaluation metrics on the LEVIR-CC dataset. Compared with closed-source methods, our approach outperforms CTM by + 0.62, + 0.75, + 1.06, + 3.07, + 0.77, + 0.81, and + 5.23 points across the seven metrics, demonstrating consistent superiority. Notably, on the S_m^* metric, our model attains a score of 82.62, surpassing CTM by 2.47 points and CD4C by 2.70 points. In comparison with open-source methods, our method leads the TACC and Chg2Cap models by 3.41 and 3.58 points, respectively, in S_m^* and achieves substantial gains across all metrics. These results strongly validate the effectiveness of the proposed method; it not only exhibits robustness against various interfering factors but also captures subtle changes in complex scenes with high sensitivity.

Table II presents the experimental results on the DUBAI-CC dataset. This dataset presents a more challenging scenario due to its limited number of samples and lower spatial resolution, demanding a stronger generalization capability from the model. Compared with existing open-source methods, our approach achieves state-of-the-art performance across all evaluation metrics. Specifically, it obtains an S_m^* score of 64.78, outperforming TACC and Chg2Cap by 9.50 and 9.34 points, respectively. However, on the same metric, our method still lags behind the closed-source model SGD-RSCCN by 2.38 points. This performance gap may stem from SGD-RSCCN's incorporation of dependency syntactic parsing, which enhances its modeling of linguistic structures. Despite this gap, our method exhibits strong robustness and generalization, highlighting its practical applicability in real-world RSICC tasks.

Fig. 5 compares model size with performance. Our method achieves superior results with a compact architecture of only 68.68 M parameters. Compared to CD4C, it reduces the parameter count by 11.56 M while improving the S_m^* score by 2.7 points, highlighting its high parameter efficiency. While SEN achieves competitive performance with just 39.9 M parameters, this comes at the cost of large-scale pretraining data and strategy. In contrast, our model is trained from scratch

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TABLE III
ABLATION STUDIES FOR EACH COMPONENT

DGDC	METI	MFA	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE$_L$	CIDEr	$S_m^\star$
✗	✓	✓	86.05 73.98	78.25 61.44	71.87 51.32	66.80 41.04	40.47 30.04	75.57 63.17	139.84 106.56	80.67 60.20
✓	✗	✓	85.01 74.98	77.05 63.44	70.49 53.51	65.17 43.58	39.73 29.41	75.18 58.67	139.44 96.17	79.88 56.96
✓	✓	✗	85.57 67.63	77.68 57.82	71.12 48.65	65.95 38.88	40.06 26.63	74.94 57.49	137.41 93.77	79.59 54.19
✓	Bi	✓	86.48 69.17	78.88 56.22	72.45 46.60	67.28 37.52	40.49 27.69	75.61 59.29	140.19 97.29	80.89 55.45
✓	✓	✓	**87.13** **75.98**	**79.71** **64.48**	**73.36** **54.68**	**68.12** **44.45**	**41.29** **32.01**	**76.53** **67.35**	**143.24** **112.14**	**82.30** **63.99**

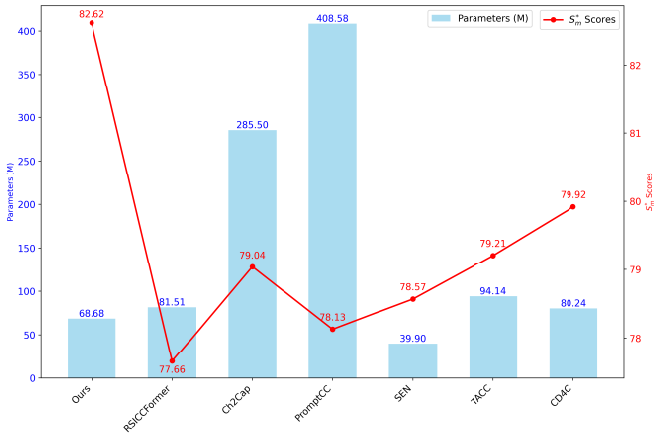


Fig. 5. Comparison of parameters and S_m^* on the LEVIR-CC dataset.

on the RSICC dataset without leveraging any external data, yet still surpasses existing methods, underscoring its practicality and effectiveness under realistic training conditions.

Furthermore, the second-stage fine-tuning—where only the DFS module is updated—yields consistent performance gains. On LEVIR-CC, all metrics except ROUGE-L improve, resulting in a 0.32-point increase in S_m^*. A similar trend is observed on DUBAI-CC, where S_m^* rises by 0.79 points. This stable improvement indicates that fine-tuning the DFS module establishes an effective feedback loop grounded in expert assessments of caption quality. Consequently, the fusion strategy can be adjusted dynamically, an optimized expert-collaboration mechanism is learned, and caption generation is significantly enhanced.

Overall, our method outperforms current state-of-the-art approaches in both performance and parameter efficiency, achieving an excellent balance between model capability and complexity. This advantage primarily stems from two key design innovations: 1) the proposed dynamic encoding mechanism effectively captures genuine change features while suppressing responses to pseudo-changes and 2) the unidirectional cross-image interaction mechanism enables the model to extract reliable and discriminative semantic change cues, enhancing both accuracy and interpretability in change captioning.

D. Ablation Studies

1) Ablation Study of Each Component: We conduct comprehensive ablation studies on the proposed DGDC, METI, and MFA modules. As shown in Table III, removing any of these components leads to a significant performance drop, verifying that the three modules work synergistically to achieve robust and high-quality change description generation.

First, compared to using standard DWC to model local dependencies, the proposed DGDC module (based on DWC) achieves consistent performance improvements across all metrics. This is because DWC struggles to effectively capture scene-dependent regions of interest, whereas DGDC dynamically generates adjustment parameters for the convolutional kernels based on difference features, thereby constructing a sample-adaptive convolutional layer that significantly enhances the model's ability to model local dependencies under diverse scenes.

Second, ablating either the METI or MFA module results in a comparable performance degradation, yet their combined effect surpasses that of the DGDC module, highlighting their pivotal role in capturing semantic changes. The METI module facilitates structured cross-image interaction through a unidirectional encoding scheme, preserving the temporal order from "before" to "after" image. In contrast, a bidirectional variant (denoted as Bi) disrupts this temporal sequence, leading to inferior performance. This comparison validates the importance of asymmetric encoding in maintaining contextual coherence, particularly for the "after" image. Meanwhile, the MFA module enables effective integration of features across multiple levels, constructing a rich, multiscale interaction space through heterogeneous fusion experts. This hierarchical aggregation is essential for handling diverse and complex visual transformations.

Furthermore, as illustrated in Fig. 6, we analyze the parameter count and computational cost (measured in FLOPs) of each component. As shown in Fig. 6(a) and (b), the DGDC, METI, and MFA modules together account for 53.05% of the total parameters but only 19.21% of the total FLOPs, demonstrating high computational efficiency. This stems from the structural design of DGDC. During inference, the cascade of adaptively weighted depthwise convolutions can be fused into a single equivalent standard convolution layer, drastically

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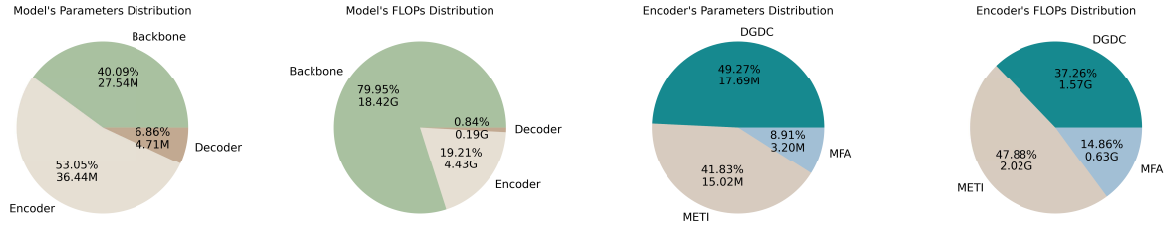


Fig. 6. Parameters and FLOPs of each component in the model.

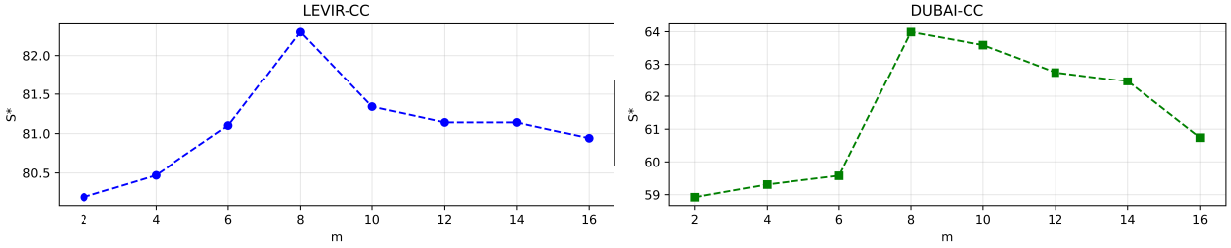


Fig. 7. Effect of different m's on the proposed DGDC module.

TABLE IV

ABLATION STUDY OF THE HETEROGENEOUS EXPERT ON THE PROPOSED METI MODULE. IN ADDITION TO PERFORMANCE METRICS, WE ALSO REPORT THE NUMBER OF PARAMETERS (M), FLOPS (G), AND INFERENCE SPEED (IMAGE PAIRS/S)

GIE	RIE	PIE	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE$_L$	CIDEr	$S_m^\star$	Params	FLOPs	FPS
✗	✓	✓	86.66 73.58	79.30 62.35	73.10 52.88	68.06 42.72	41.08 30.12	76.16 64.30	140.98 106.06	81.57 60.80	31.71	3.81	614
✓	✗	✓	85.72 73.05	78.10 61.31	71.47 50.99	66.07 41.40	40.19 29.12	75.61 62.22	140.62 101.05	80.62 58.45	33.24	4.11	598
✓	✓	✗	85.88 75.96	78.13 63.54	71.67 53.24	66.60 42.72	40.29 31.82	75.39 65.55	139.83 109.62	80.53 62.43	34.07	4.27	591
✓	✓	✓	**87.13** **75.98**	**79.71** **64.48**	**73.36** **54.68**	**68.12** **44.45**	**41.29** **32.01**	**76.53** **67.35**	**143.24** **112.14**	**82.30** **63.99**	36.44	4.43	583

reducing runtime overhead without compromising representational expressiveness.

2) Ablation Study of DGDC: The proposed DGDC module comprises m structurally identical DWC layers. The choice of m is critical, as it determines the model's ability to capture local dependencies across different semantic levels and spatial perspectives. However, an excessively large m increases model complexity, which may lead to overfitting—particularly on RSICC datasets with limited training samples. To investigate this tradeoff, we conduct ablation studies with different values of m (specifically, $m = [2, 4, 6, 8, 10, 12, 16]$), and the results are shown in Fig. 7. It can be observed that, on both datasets, the model performance reaches a saturation point and achieves the best accuracy when $m = 8$.

3) Ablation Study of METI: The proposed METI module comprises three heterogeneous experts: GIE, RIE, and PIE, which facilitate cross-image interactions with distinct functionalities. Subsequently, a DFS module computes the contribution of each expert to enable adaptive feature fusion.

First, to validate the effectiveness of each expert, we evaluated their specific impact on model performance. As shown in Table IV, removing GIE results in a 0.73/3.19 drop in the $S_m^\star$ metric, alongside significant reductions in ROUGE$_L$ and CIDEr scores. This indicates that GIE plays a crucial role in modeling macrosemantic relationships across image sequences, enhancing the coherence and semantic integrity of generated descriptions. In contrast, removing RIE and PIE leads to more pronounced performance degradation, with $S_m^\star$ metric drops of 1.68/5.54 and 1.77/1.56, respectively. Specifically, the absence of RIE primarily affects BLEU-4 and METEOR scores, demonstrating its role in improving syntactic complexity by modeling region-level visual–semantic relationships (e.g., spatial structures between objects). Meanwhile, the failure of PIE causes notable fluctuations in ROUGE$_L$ and CIDEr metrics, further verifying its fidelity in extracting discriminative features at the keypoint level. Notably, although RIE and PIE exhibit similar impacts on the $S_m^\star$ metric, their failure mechanisms differ fundamentally. RIE mainly alleviates local semantic ambiguities, and its absence often leads to syntactic errors, whereas PIE focuses on capturing fine-grained discriminative features, and its removal tends to omit content-related information.

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TABLE V
ABLATION STUDY OF THE DFS ON THE PROPOSED METI MODULE

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE$_L$	CIDEr	$S_m^\star$
Mean	86.04 74.61	78.57 62.19	72.33 51.42	67.36 42.59	40.59 29.86	75.44 60.58	140.44 100.47	80.96 58.37
MLP	85.80 72.79	78.31 62.79	71.97 53.13	66.86 43.52	40.64 29.13	75.56 58.62	139.87 98.54	80.73 57.46
DFS	**87.13** **75.98**	**79.71** **64.48**	**73.36** **54.68**	**68.12** **44.45**	**41.29** **32.01**	**76.53** **67.35**	**143.24** **112.14**	**82.30** **63.99**

TABLE VI
ABLATION STUDY OF THE PROPOSED MFA

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE$_L$	CIDEr	$S_m^\star$
Last	85.57 67.63	77.68 57.82	71.12 48.65	65.95 38.88	40.06 26.63	74.94 57.49	137.41 93.77	79.59 54.19
Mean	85.61 72.38	77.77 60.62	71.19 50.78	65.95 41.55	39.98 29.09	74.99 61.99	138.21 100.72	79.78 58.34
MLP	85.52 **76.71**	78.01 **65.27**	71.83 **55.17**	66.93 **44.78**	40.50 31.25	75.62 62.50	139.97 105.72	80.76 61.06
MFA	**87.13** 75.98	**79.71** 64.48	**73.36** 54.68	**68.12** 44.45	**41.29** **32.01**	**76.53** **67.35**	**143.24** **112.14**	**82.30** **63.99**

From the perspective of computational efficiency, the complexity of each expert module is positively correlated with its receptive field size. Specifically, the GIE module, which focuses on global interaction, has the largest number of parameters (4.73 M) and the highest computational cost (0.62 GFLOPs), significantly exceeding those of the RIE (3.2 M parameters and 0.32 GFLOPs) and PIE (2.37 M parameters and 0.16 GFLOPs), which are dedicated to local detail modeling. This difference in complexity directly translates into inference speed; removing the GIE module, which imposes the heaviest computational burden, increases the model's inference frame rate from 583 to 614 frames/s, achieving a notable speed improvement. This indicates that in applications with high demands on inference efficiency, the overall efficiency can be optimized by rationally pruning expert modules through a tradeoff between model performance and computational cost.

Second, to assess the effectiveness of the DFS module, we designed two baseline methods: parameter-free mean pooling and learnable MLP-based dimensionality reduction. As shown in Table V, the parameter-free method outperforms the learnable method, indicating that averaging operations better preserve the original semantic distribution of heterogeneous features, while parameterized fusion may disrupt the semantic consistency of expert representation spaces. Additionally, both static fusion strategies (mean pooling and MLP) perform significantly worse than DFS across all metrics, suggesting that fixed weight allocation struggles to adapt to the dynamic variations in expert contributions under different scenarios (e.g., the importance of global, regional, and point-level interactions varies with input), thereby limiting the model's adaptability to complex semantic changes. In contrast, DFS dynamically integrates expert outputs based on different inputs, forming differentiated interaction patterns and significantly enhancing the model's adaptability to diverse semantic variations.

In summary, the removal of any expert significantly degrades model performance, validating the functional complementarity of GIE, RIE, and PIE. These components collaborate within a multilevel encoding space and, combined with DFS, construct a flexible temporal–semantic interaction framework. This framework dynamically enhances the weights of critical experts while effectively suppressing redundant information, thereby improving the semantic completeness and robustness of feature representations in complex scenarios.

4) Ablation Study of MFA: The MFA module is designed to process multilevel features extracted by the DAE and transform them into semantically enriched representations that are more interpretable for the language model. As a pivotal component of the overall framework, its structural design directly influences performance. To systematically evaluate the impact of different architectural choices on the MFA module's effectiveness, we compare three alternative methods: 1) *last*, which utilizes only the output features from the final layer of the encoder; 2) *mean*, which aggregates multilevel features by averaging across layers; and 3) MLP, which concatenates multilevel features along the feature dimension and subsequently reduces dimensionality using an MLP. As shown in Table VI, these baseline methods achieve suboptimal results. Given that the multilevel features correspond to outputs from different layers of the encoder, they inherently exhibit sequential dependency. To exploit this characteristic, we propose a progressive feature integration strategy that incrementally combines features while preserving their hierarchical relationships. Experimental results confirm that the

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TABLE VII
EFFECT OF DIFFERENT DEPTHS

N	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE$_{LL}$	CIDEr	$S_m^\star$
1	84.64 69.61	76.55 57.97	69.83 48.33	64.51 40.63	39.57 28.41	74.63 59.22	136.65 93.33	78.84 55.40
2	85.66 71.45	77.90 60.43	71.43 51.05	66.26 42.17	40.10 30.65	75.25 64.83	139.16 105.13	80.19 60.69
3	**87.13** 75.98	**79.71** **64.48**	**73.36** 54.68	**68.12** 44.45	**41.29** **32.01**	**76.53** **67.35**	**143.24** **112.14**	**82.30** **63.99**
4	85.93 **76.01**	78.46 64.18	72.08 **54.84**	67.04 **45.01**	40.55 31.12	75.59 65.14	140.27 105.17	80.86 61.61

TABLE VIII
EFFECT OF DIFFERENT KERNEL SIZES

K	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE$_L$	CIDEr	$S_m^\star$
3	85.58 72.64	77.90 60.78	71.28 51.36	66.10 43.32	40.08 30.21	75.09 61.33	138.11 100.90	79.85 58.94
5	85.36 72.98	77.60 62.36	71.14 52.36	65.95 41.88	40.40 29.22	75.27 61.41	139.32 108.50	80.24 60.25
7	84.24 71.67	76.13 60.85	69.71 51.35	64.81 40.34	39.38 30.18	73.88 64.18	134.54 105.65	78.15 60.09
$2l+1$	**87.13** **75.98**	**79.71** **64.48**	**73.36** **54.68**	**68.12** **44.45**	**41.29** **32.01**	**76.53** **67.35**	**143.24** **112.14**	**82.30** **63.99**

proposed MFA module significantly enhances the quality of generated captions, thereby validating its efficacy.

5) Ablation Study on Depths: In our model, the encoder first generates a set of change features, which are then progressively refined through the proposed MFA module, ultimately yielding more accurate change representations. In this process, the depth N of the encoder is consistent with the depth of the MFA module. Generally, if N is too small, it limits the model's ability to perform semantic modeling within images and change perception across images. Conversely, if N is too large, it may lead to overfitting under limited data conditions. For instance, the model may struggle to effectively learn the large-scale interaction space constructed by the METI module, while the progressive feature encoding capability of the MFA module could also be constrained. To investigate the impact of different values of N on model performance, we set $N = \{1, 2, 3, 4\}$ and conducted experiments, with results shown in Table VII. It can be observed that as N increases, the model performance improves gradually, reaching its optimal performance at $N = 3$.

6) Ablation Study on Kernel Size: The proposed model employs a large number of convolutional layers, where the kernel size k plays a crucial role in balancing local detail extraction and receptive field control. We evaluate four configurations: $k \in \{3, 5, 7, 2l + 1\}$, with l denoting the layer index. Experimental results show that increasing k from 3 to 5 improves performance, indicating that a moderately enlarged receptive field benefits context modeling. However, further increasing k to 7 leads to performance degradation: on the LEVIR-CC dataset, the CIDEr score drops by 3.78, and on the DUBAI-CC dataset, it decreases by 2.85. This suggests that an excessively large convolutional kernel may oversmooth critical change-related features. In contrast, the layer-adaptive configuration $k = 2l + 1$ achieves the best results across all metrics. This design progressively expands the receptive field: small kernels in shallow layers preserve fine details, while large kernels in deeper layers capture high-level semantic context. This hierarchical approach achieves an optimal balance between local and global feature representations, enhancing the discriminability among multilevel features generated by DAE, thereby providing more effective support for fusion in the MFA module.

E. Qualitative Analysis

To validate the effectiveness of the proposed method, we selected three representative sample sets from the LEVIR-CC dataset for qualitative analysis. The first set comprises pseudo-change cases induced by interference, designed to evaluate the model's stability in visual representation learning. The second set includes complex scene samples, intended to assess the model's capability to distinguish between unchanged and semantically changed regions. The third set focuses on edge-region samples, aimed at examining the model's precision in localizing subtle changes. Meanwhile, we selected a representative set of samples from the DUBAI-CC dataset. In the experiments, RSICCFormer and Chg2Cap were adopted as baseline models, and the results are shown in Fig. 8, where green annotations indicate correct descriptions, orange annotations denote vague descriptions, and red annotations represent incorrect descriptions. Additionally, we present the contribution distribution of each expert in the METI module.

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Fig. 8. Qualitative comparison of generated captions on the LEVIR-CC and DUBAI-CC datasets. Green, red, and brown denote correct, incorrect, and ambiguous words, respectively. GT indicates the ground-truth sentence.

In the first set of samples, significant lighting differences exist between the bi-temporal images. As a result, both RSICCFormer and Chg2Cap suffer from semantic drift, mistakenly interpreting lighting variations as semantic changes. In contrast, our method effectively suppresses the impact of lighting discrepancies, preserving the capability to extract stable semantic representations.

In the second set of samples, RSICCFormer fails to distinguish between changed and unchanged regions, leading to descriptions that are highly inconsistent with the actual scene. While Chg2Cap mitigates localization errors by employing vague spatial descriptors (e.g., "on the bareland" and "along the road"), it compromises descriptive accuracy. Our method, however, efficiently suppresses responses from nonsemantic change regions and generates precise spatial descriptions such as "at the top of the scene," demonstrating superior semantic discrimination performance.

In the third set of samples, the changed regions occupy only 5% of the total image area and exhibit minimal visual differences. Nevertheless, our method accurately localizes the edge-located change regions, whereas the baseline methods suffer from substantial false negatives or false positives. This demonstrates that our method not only maintains semantic correctness but also delivers interpretable and accurate localization information.

In the fourth set of samples, low-resolution images under illumination interference are highly prone to cause the model to generate misclassifications or hallucination phenomena, such as incorrectly recognizing "a road" as "a park" or identifying changed regions as unchanged. In contrast, our method still produces accurate descriptions, further validating its stability under complex conditions.

To further examine the usage patterns across expert modules, we statistically analyzed and visualized the dominant scenes for each expert, defined as regions in the interaction space exhibiting the highest average gating weights (Fig. 9). In GIE-dominant scenes (first row), the model is primarily activated by no-change pairs or highly complex changes. In no-change scenarios, the absence of salient differences necessitates reliance on the global context to verify cross-temporal semantic consistency. In complex-change cases—e.g., "some buildings are removed while new houses and roads are constructed"—the transformation involves multiple heterogeneous objects, and global contextual cues are essential for accurate localization and interpretation of the real changed regions. In RIE-dominant scenes (second row), representative instances are localized and semantically unambiguous additions, such as a single new building emerging in a forested area. Because the pre-change image provides no evidence of the structure, the change is spatially confined yet visually salient. In such cases, RIE's focus on local differences demonstrates both sufficiency and computational efficiency. In PIE-dominant scenes (third row), typical examples include large-scale land-cover transformations, such as the conversion of an entire forest into a residential area. These changes are substantial at both the pixel and semantic levels, and PIE's point-wise interaction mechanism effectively models the state transitions of individual pixels. Although the spatial extent of the change is significantly larger than in RIE-dominated cases, the underlying task—detecting newly emerged structures—is conceptually similar; this similarity explains the nonnegligible contribution of RIE in these scenarios.

In summary, our method demonstrates excellent stability and discriminative capability across various scenarios, including no-change, partial-change, and complex semantic change cases. This advantage stems from the synergistic design of the dynamic encoding mechanism and the asymmetric interaction architecture, enabling the model to adaptively select optimal

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Fig. 9. Visualized results on the LEVIR-CC dataset under expert-dominant scenarios, where each expert exhibits higher average weights in the interaction space. The three rows sequentially present the dominant scenarios for GIE, RIE, and PIE. Each row includes the average weight distribution of the corresponding expert and the associated ground-truth caption.

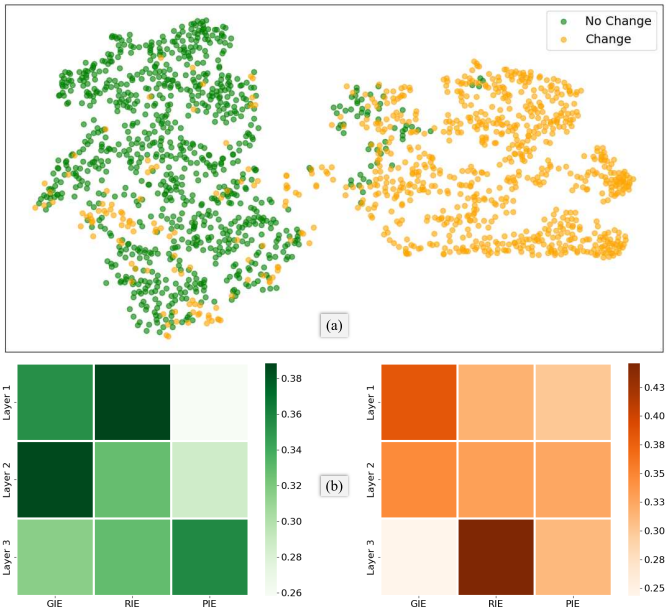


Fig. 10. Visualization of expert results on the LEVIR-CC dataset. (a) Clustering visualization using the t-SNE algorithm [50]. (b) Distribution of expert preferences in the three-level interaction space under scenarios with and without semantic changes.

feature interaction pathways, accurately interpret real change patterns, and generate high-quality change captions.

F. Expert Visualization

Ideally, the DFS module should adaptively select an appropriate interaction mode based on the input sample by dynamically adjusting the weights assigned to individual expert modules. To validate this capability, we analyze the weight distribution within the DFS interaction space and employ t-SNE [50] for clustering visualization, revealing expert preference patterns under both unchanged and semantically changed conditions.

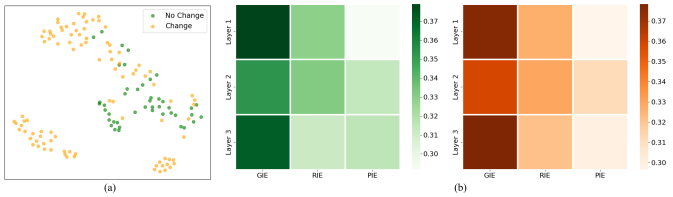


Fig. 11. Visualization of expert results on the DUBAI-CC dataset. (a) Clustering visualization using the t-SNE algorithm [50]. (b) Distribution of expert preferences in the three-level interaction space under scenarios with and without semantic changes.

Results on the LEVIR-CC dataset are shown in Fig. 10. As illustrated in Fig. 10(a), the model exhibits clear clustering separation between unchanged samples (in green) and semantically changed samples (in orange), demonstrating its ability to effectively distinguish between these two semantic categories and highlighting its strong semantic understanding and feature abstraction capabilities. Moreover, Fig. 10(b) reveals the model's layer-wise expert weighting preferences. In unchanged scenarios, the first layer predominantly activates RIE, the second layer favors GIE, and the third layer shows a more balanced allocation across experts. In contrast, under semantic changes, the first layer shifts toward GIE, the second layer adopts a more balanced strategy, and the third layer places greater emphasis on RIE.

On DUBAI-CC (Fig. 11), the embeddings of changed and unchanged pairs are clearly separated. The unchanged samples further form three compact subclusters, a pattern we attribute to the greater textual diversity in unchanged scenes—24 unique captions in the test set compared to only five in LEVIR-CC. Across hierarchical expert preferences, the first two layers exhibit stable weight distributions, while the third layer shows notable divergence. For unchanged pairs, RIE and PIE receive nearly equal weights, indicating balanced contributions from both experts. In contrast, when semantic changes occur, the weight assigned to RIE increases significantly, highlighting its enhanced role in capturing change-related semantics.

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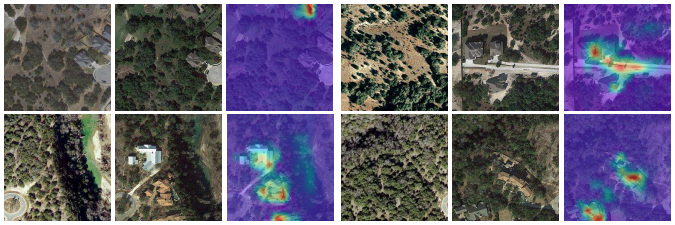


Fig. 12. Visualization of the text-to-image attention maps of our method on the LEVIR-CC dataset.

The above results demonstrate that the proposed method exhibits differentiated expert selection patterns across various scenarios, indicating that the model has learned to adaptively adjust its computational paths according to input semantics. Notably, during training, no explicit constraints or prior knowledge regarding scene types were introduced. Therefore, this adaptive computation capability is acquired indirectly through the pure descriptive generation task. This not only further validates the effectiveness of the method but also preserves flexibility for exploration across different scenarios.

Nonetheless, certain limitations persist. For instance, the model does not achieve perfect discriminative processing, as evidenced by a small number of misclustered samples in the visualization. This may be attributed to the model's limited generalization capacity in handling complex scenes, indicating that future work could focus on enhancing robustness and discriminative learning in such contexts.

G. Attention Visualization

Fig. 12 illustrates the visualization of the attention weights in the last layer of the decoder. As shown in Fig. 12, our method accurately focuses on the regions of interest corresponding to changes in the image pairs. Specifically, for the first image pair, our method captures subtle changes located in the top-right corner, wherein the area of the changed object constitutes less than 5% of the entire image. Multiple objects undergo changes in subsequent image pairs. Even when these objects are discontinuous, as demonstrated in image pair 4, our method identifies them individually. This indicates that the proposed model effectively mitigates the influence of irrelevant visual semantics, thereby accurately identifying the objects that have changed.

V. Conclusion

In this article, we propose the DAE, which introduces two key innovations: a dynamic encoding mechanism and an asymmetric processing architecture. On the one hand, input-driven adaptive feature enhancement is achieved through a DGDC module. On the other hand, an METI module is designed to operate exclusively on the "after" branch, enabling diverse semantic perception while preserving temporal coherence. This module is further integrated with a dynamic weight fusion strategy to improve the model's robustness in complex scenarios. Consequently, the combination of input-adaptive encoding and structural asymmetry enhances the model's ability to capture change semantics effectively. To further refine feature representations, we introduce an MFA that iteratively integrates multilevel features to generate structurally coherent and semantically enriched representations of change. Extensive experiments are conducted on two widely used RSICC datasets, demonstrating the effectiveness of the proposed method and achieving state-of-the-art performance.

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